

Visualizing Mobility of Public Transportation System

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Contents

- 1. Introduction
- 2. Related Work
- 3. Our Visual Analytics System
- 4. Case Studies
- 5. Conclusion and Future Work

Contents

- **1. Introduction**
- 2. Related Work
- 3. Our Visual Analytics System
- 4. Case Studies
- 5. Conclusion and Future Work

1.1 Public Transportation Systems

- Public transportation systems (PTS) play an important role in modern cities:
 - ✓ **Economically:** reduce the overall travel cost for the entire city
 - ✓ **Socially:** ensure all residents are able to travel
 - ✓ **Environmentally:** save more energy and reduce air pollution

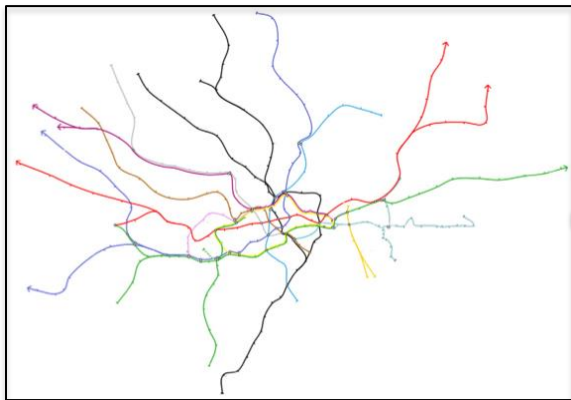


1.2 PTS Mobility

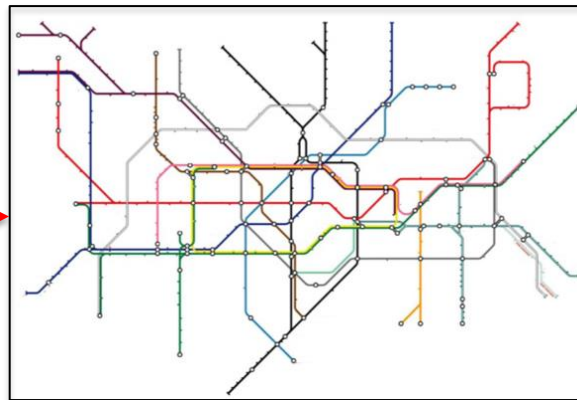
- PTS mobility refers to physical movement, measured by trips, distances and speeds through PTS [Litman 2008].
- Studying PTS mobility can:
 - ✓ indicate the efficiency of a PTS
 - ✓ assist individuals plan routes
 - ✓ reflect transportation and city design
 - ✓ etc.

1.3 Visualizing PTS

- Visualizing PTS can help people in locating position, navigating the network and wayfindings [Bartram 1980, Meilinger et al. 2007]



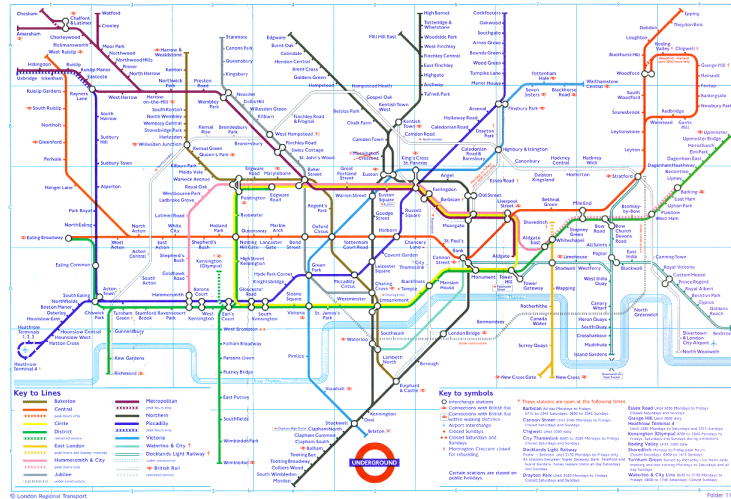
London Metro: True Map



Tube Map [since: Harry Beck 1931]

1.3 Visualizing PTS: Challenge 1

- **Complex network:** hundreds of subway stations even excluding bus stops



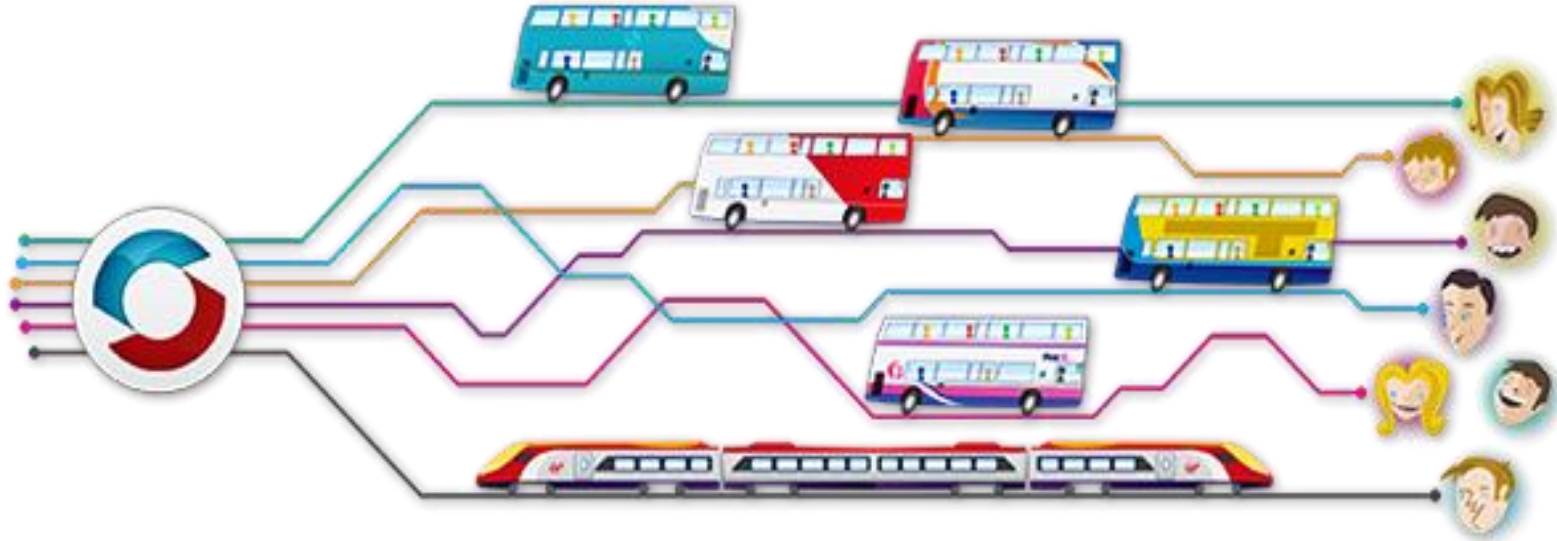
London subway (~270 stations)



New York subway (~460 stations)

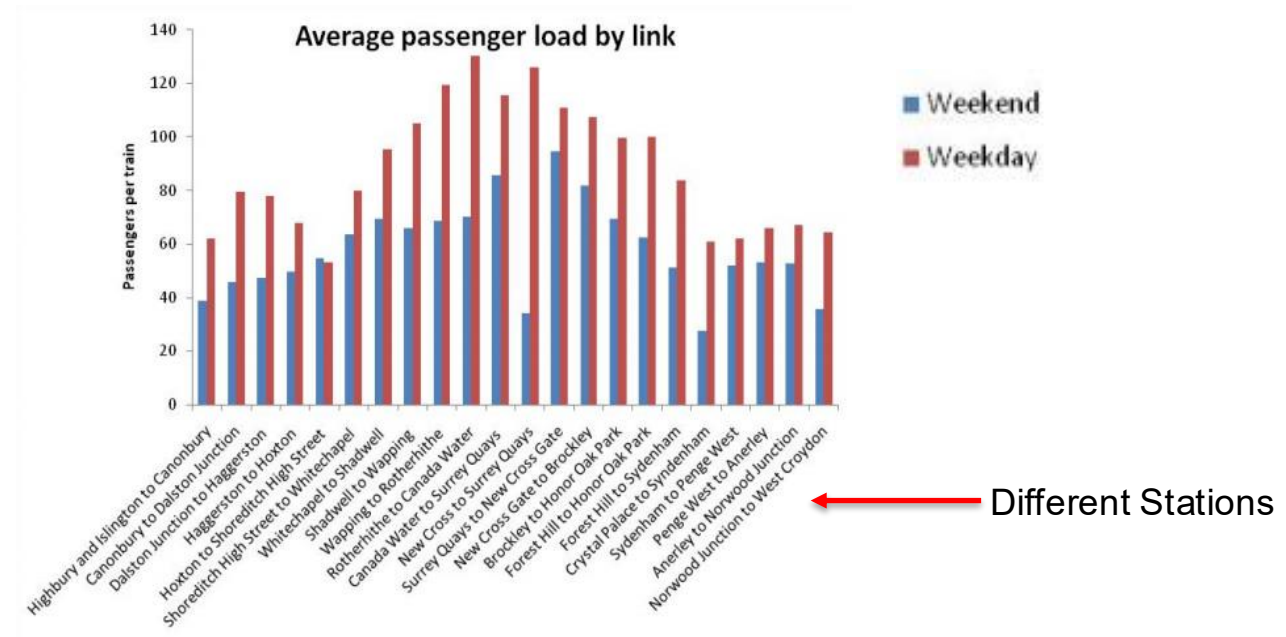
1.3 Visualizing PTS: Challenge 2

- **Multi-modes transport:** metro, tram, bus, walk, etc.



1.3 Visualizing PTS: Challenge 3

- **Spatial and temporal patterns:** traffic conditions vary dynamically with both time and space



1.4 Singapore Public Transportation Data

- We studied PTS by extracting mobility information from:
 - ✓ **Transportation network:** 48k nodes, 80k edges
 - ✓ **EZLink data:** 5.3m trip records per day
 - ✓ **Transit schedule:** transit services, vehicles leaving & arriving times at stops



Singapore public transport network

1.4 Singapore Public Transportation Data

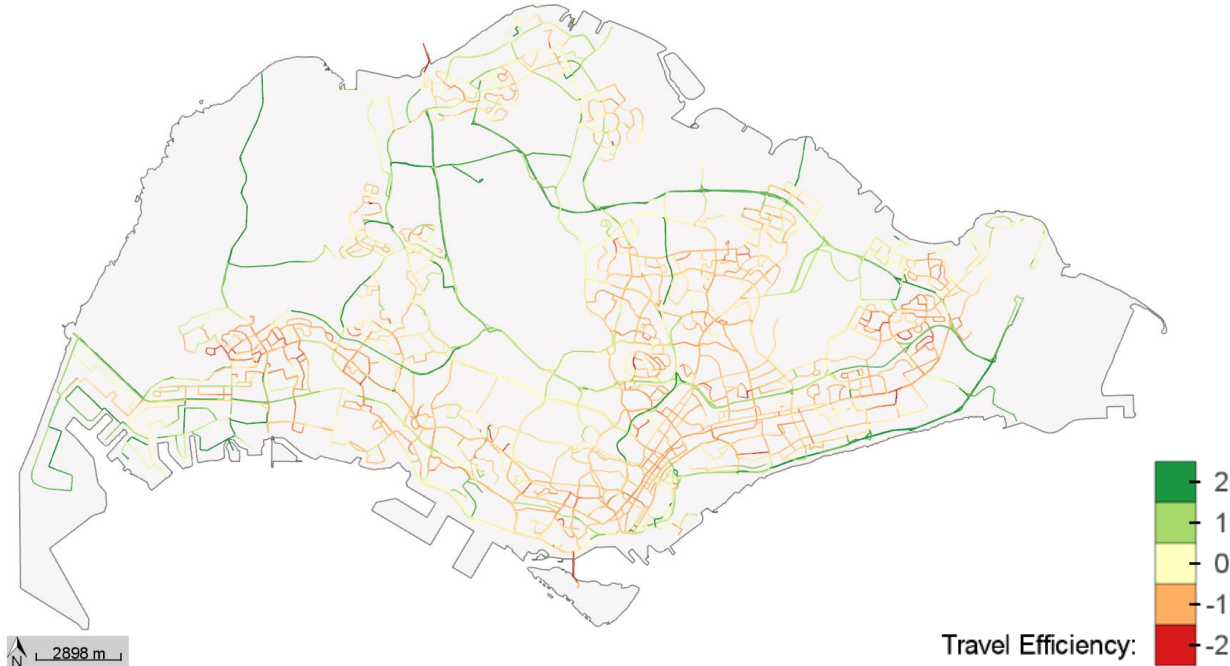
➤ In Singapore, passengers tap-in/out EZLink cards to enter/exit public transportation. For each record:

- ✓ **Journey ID**
- ✓ **Anonymous Card ID**
- ✓ **Passenger Type:** e.g., Adult, Senior, Child
- ✓ **Travel Mode:** Bus or Train
- ✓ **Service Number:** e.g., 199 for Bus, NULL for Train
- ✓ **Boarding/Alighting Stops**
- ✓ **Ride Start Data**
- ✓ **Ride Start Time**
- ✓ **Ride Distance:** 16.8km
- ✓ **Ride time:** 43.45 min
- ✓ **Fare Paid:** S\$1.49



1.4 Singapore Public Transportation Data

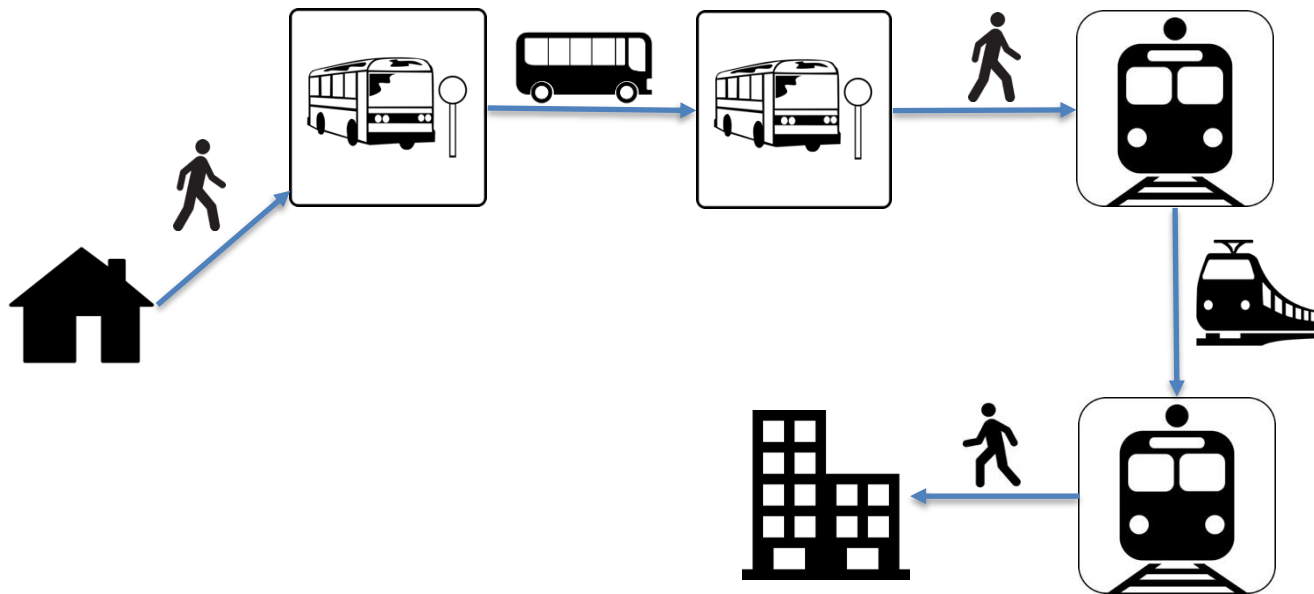
- In the data-preprocessing stage, we model and extract high-level mobility-related factors from the raw data.



Road network
travel efficiency
map at a certain
time moment

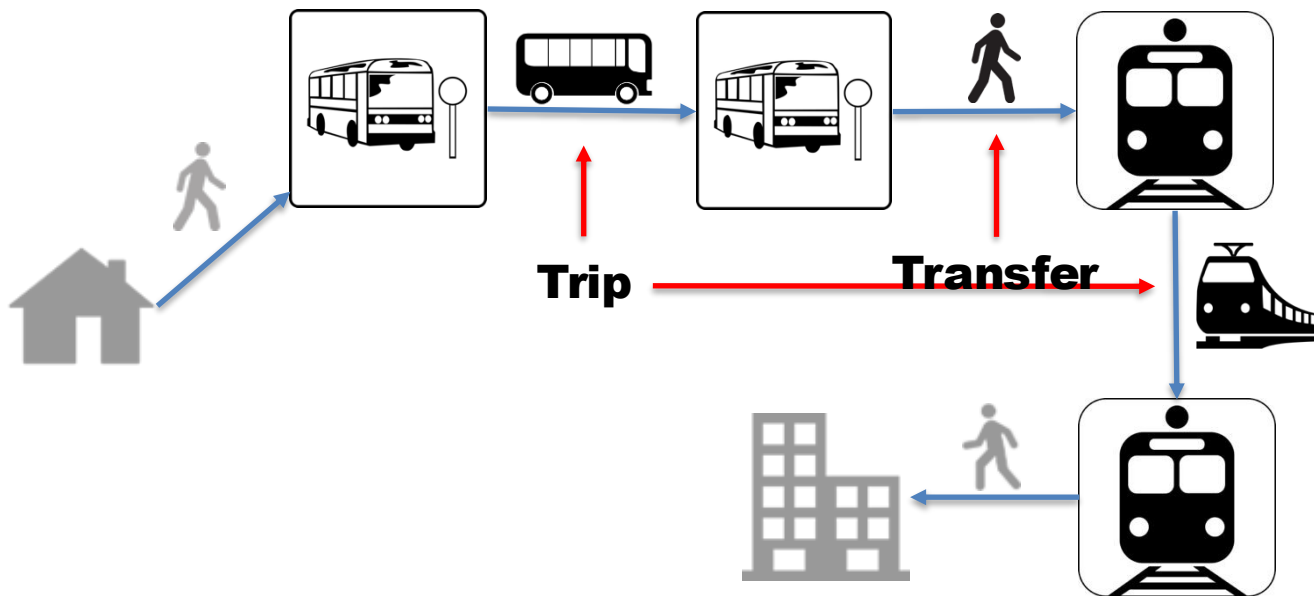
1.5 PTS Mobility Model

- An example of typical public transportation journey



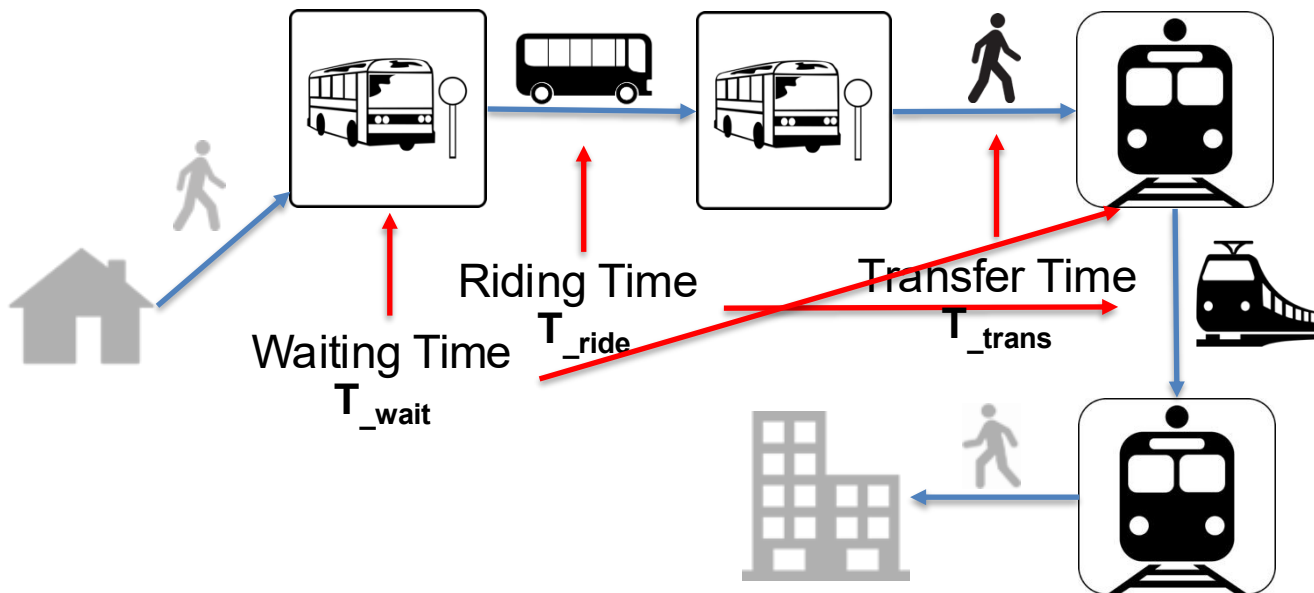
1.5 PTS Mobility Model

- We do not consider the initial and final walks



1.5 PTS Mobility Model

➤ The travel time can be decomposed into:



1.5 PTS Mobility Model

- We construct a PTS mobility model for a time-efficient journey with **n** trips and **n-1** transfers:

$$T_{journey} = \sum_{i=1}^n [T_{wait_i} + T_{ride_i}] + \sum_{i=1}^{n-1} T_{trans_i}, \quad n \geq 1$$

 **n trips**  **n-1 transfers**

Contents

- 1. Introduction
- **2. Related Work**
- 3. Our Visual Analytics System
- 4. Case Studies
- 5. Conclusion and Future Work

2.1 Transit Timetable

- Timetable lists the ***scheduled*** arrival and departure times of a service from specified locations.
- However, timetable
 - ✓ is not intuitive
 - ✓ cannot reflect the real traffic condition
 - ✓ does not tell the time-efficient routes



The image shows a yellow and blue transit timetable board from Zurich, Switzerland. The board is titled "Abfahrt Bahnhof Zürich Tiefenbrunnen" and "11 December 2005 - 9 December 2006". It displays departure times for various train lines, including S1, S2, S3, S4, S5, S6, S7, S8, S9, S10, S11, S12, S13, S14, S15, S16, S17, S18, S19, S20, S21, S22, S23, S24, S25, S26, S27, S28, S29, S30, S31, S32, S33, S34, S35, S36, S37, S38, S39, S40, S41, S42, S43, S44, S45, S46, S47, S48, S49, S50, S51, S52, S53, S54, S55, S56, S57, S58, S59, S60, S61, S62, S63, S64, S65, S66, S67, S68, S69, S70, S71, S72, S73, S74, S75, S76, S77, S78, S79, S80, S81, S82, S83, S84, S85, S86, S87, S88, S89, S90, S91, S92, S93, S94, S95, S96, S97, S98, S99, S100. The board is organized into columns for different train lines and rows for departure times.

A timetable in Zurich. [source: wikipedia]



The image shows a digital transit timetable board from Yokohama, Japan. The board is titled "みなとみらい線標準時刻表" and "Train Departures of Minatomirai Line". It displays departure times for various train lines, including M1, M2, M3, M4, M5, M6, M7, M8, M9, M10, M11, M12, M13, M14, M15, M16, M17, M18, M19, M20, M21, M22, M23, M24, M25, M26, M27, M28, M29, M30, M31, M32, M33, M34, M35, M36, M37, M38, M39, M40, M41, M42, M43, M44, M45, M46, M47, M48, M49, M50, M51, M52, M53, M54, M55, M56, M57, M58, M59, M60, M61, M62, M63, M64, M65, M66, M67, M68, M69, M70, M71, M72, M73, M74, M75, M76, M77, M78, M79, M80, M81, M82, M83, M84, M85, M86, M87, M88, M89, M90, M91, M92, M93, M94, M95, M96, M97, M98, M99, M100. The board is organized into columns for different train lines and rows for departure times. Below the timetable, there is a map of the Minatomirai Line.

A timetable in Yokohama, Japan. [source: wikipedia]

2.2 Schematic Maps

- Schematic maps overview the PTS network's essential topological relations across stops [Garland 1994]



Singapore MRT map [source: wikipedia]

2.2 Schematic Maps

- Schematic maps help passengers navigate the network and identify paths [Bartram 1980, Meilinger et al. 2007]



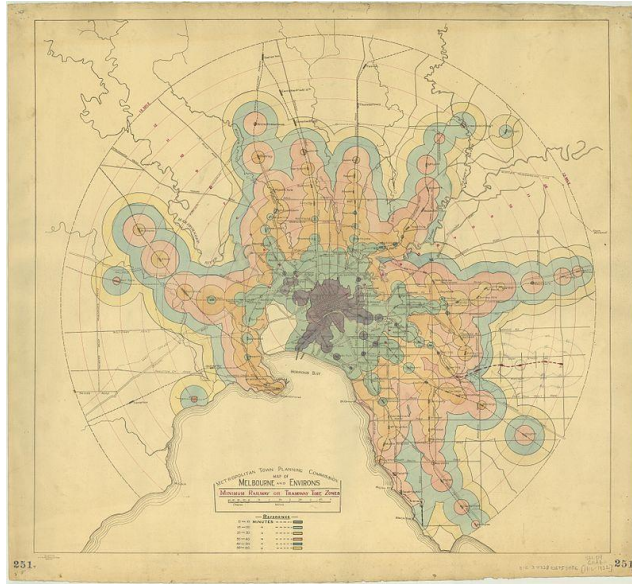
2.2 Schematic Maps

- However, schematic maps cannot tell any mobility information, e.g., travel time from origin to destination

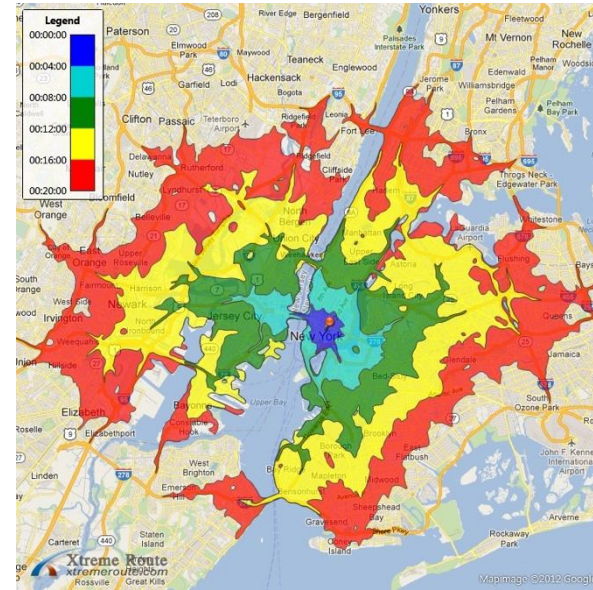


2.3 Isochrone Maps

- Isochrone maps employ countour lines/colors to present areas of equal travel time from a given starting location



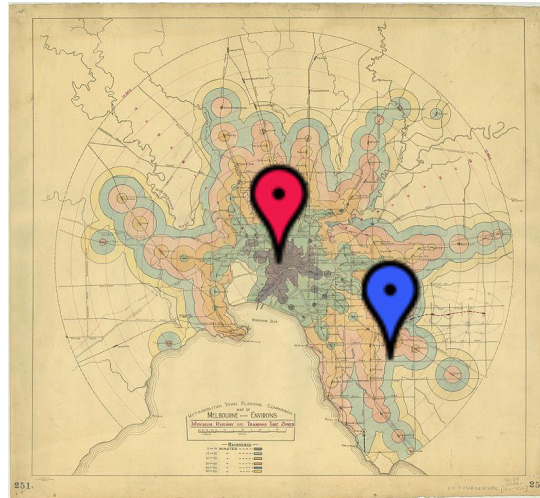
Isochrone map of Melbourne travel times to the city center by tram or train. [source: wikipedia]



Drivetime polygons in New York with 4 minute intervals. [source: Xtreme Route]

2.3 Isochrone Maps

- Isochrone maps preserve geographical information by overlaying travel time information on maps.
- However, they cannot reveal ***detailed*** travel time information and routes from the starting location to destinations.



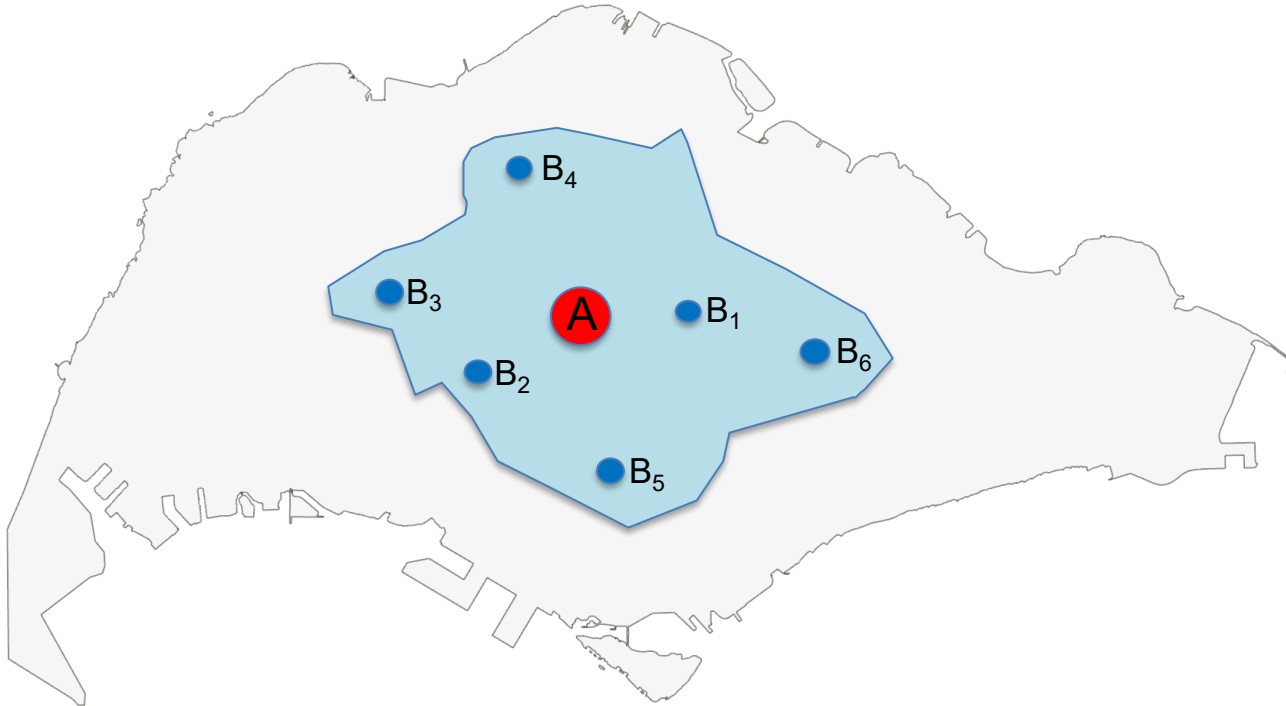
What is the exact travel time and routes?

Contents

- 1. Introduction
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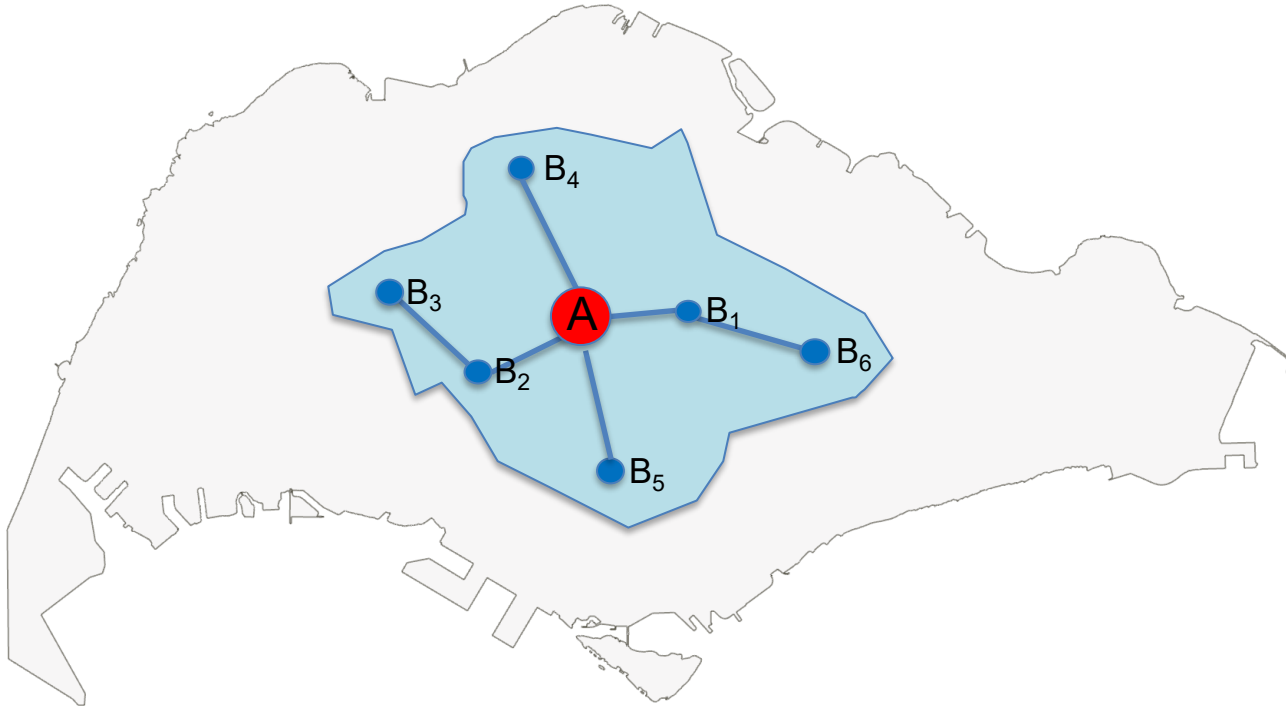
3.1 Analytical Tasks

- Task 1: What destinations can be reached from A within a given time period?



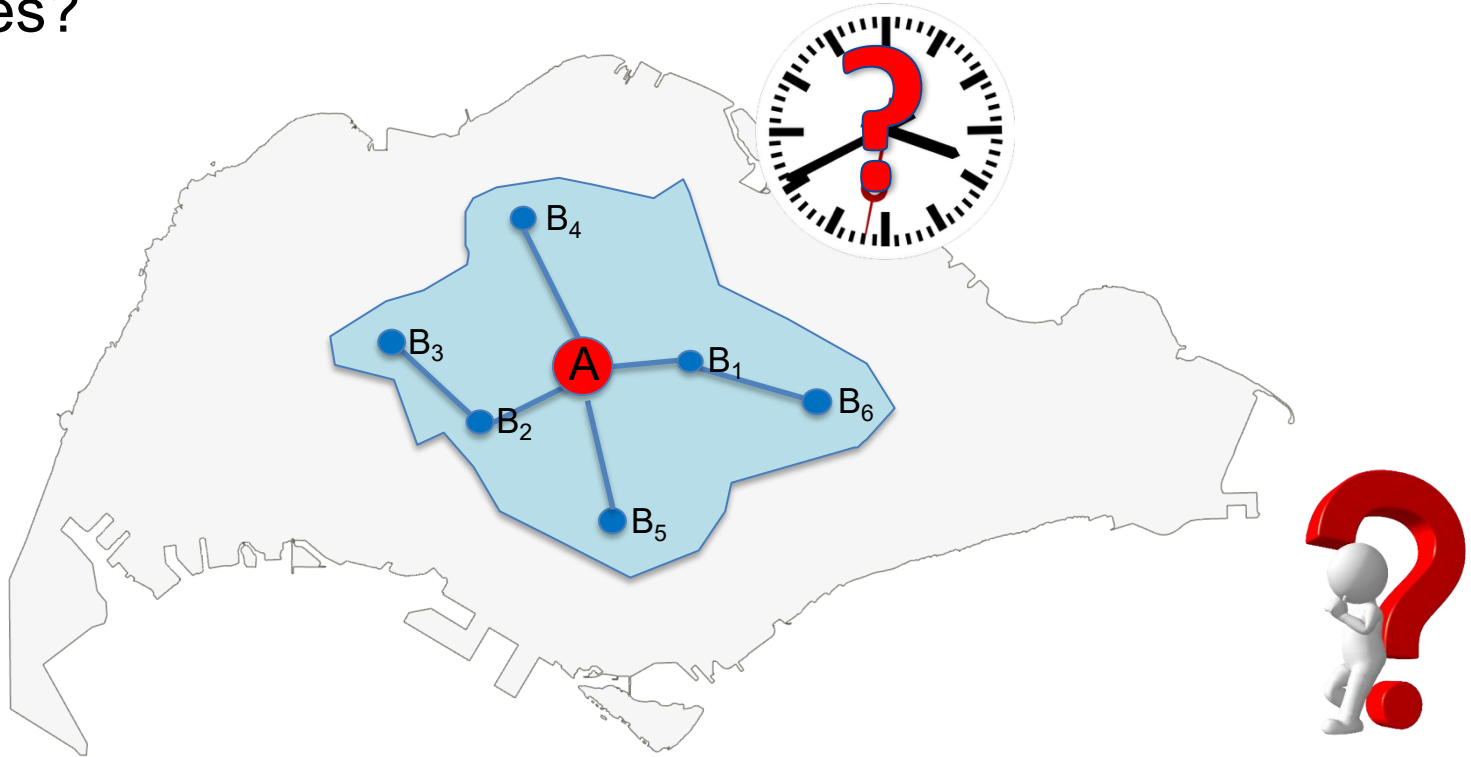
3.1 Analytical Tasks

- Task 2: What are the time-efficient routes from A to the destinations?



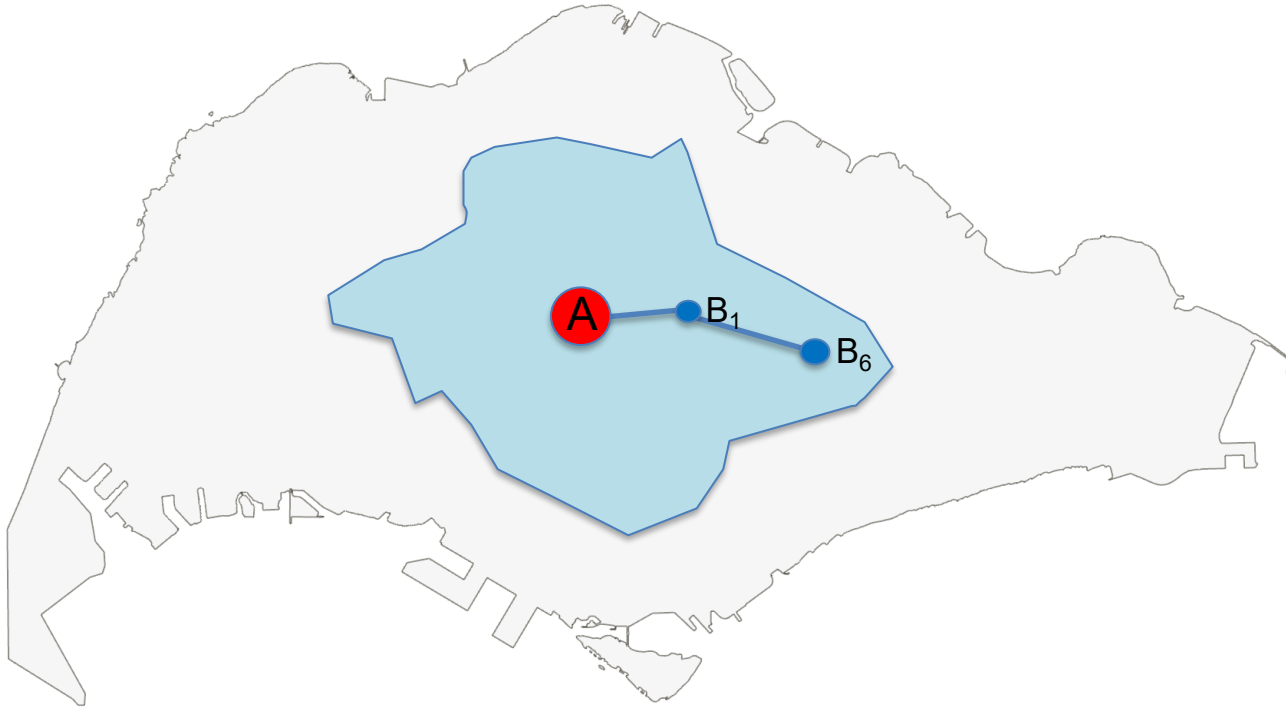
3.1 Analytical Tasks

- Task 3: What are the travel time and efficiency of these routes?



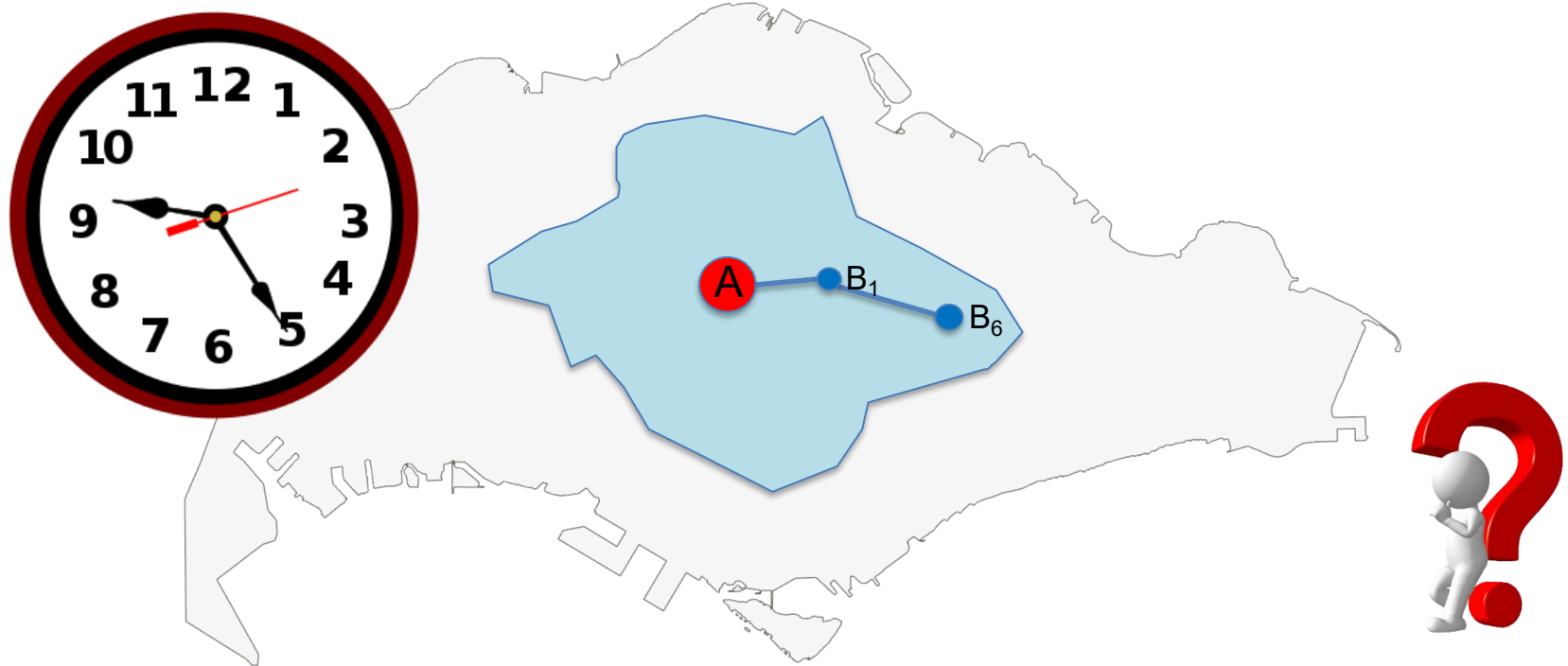
3.1 Analytical Tasks

- Task 4: Given a specific destination, what is the detailed mobility information (waiting, riding, and transfer)?



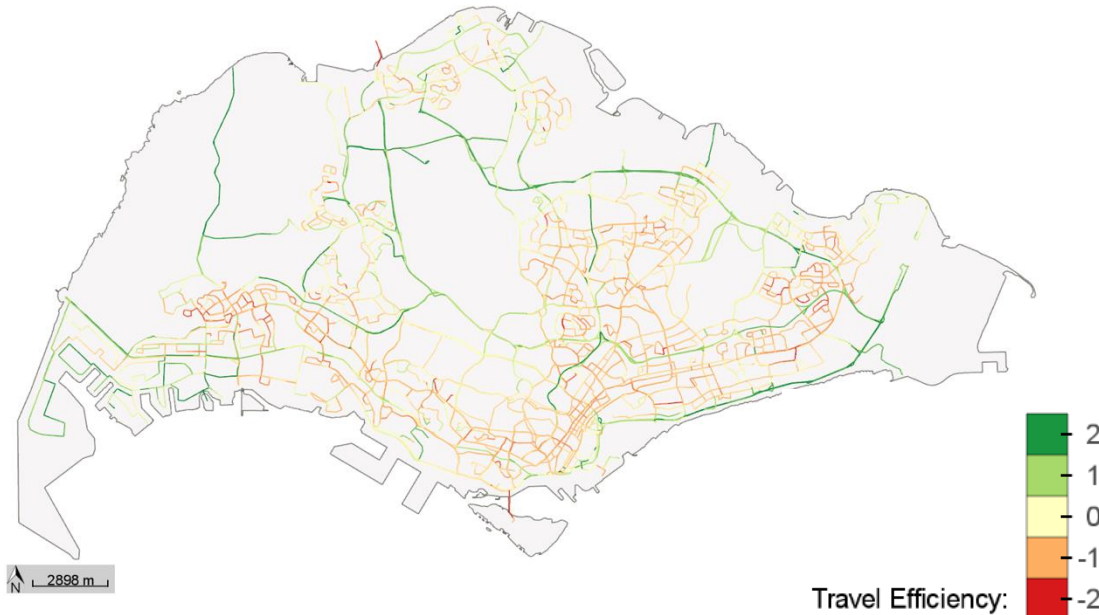
3.1 Analytical Tasks

- Task 5: What is the temporal variation (over one day) of these mobility factors?



3.2 Data Pre-processing

- In the data-preprocessing stage, we extract the mobility factors from *EZLink data*, *transportation network*, and *transit schedule*.



Road network
travel efficiency
map at a certain
time moment

3.3 Isochrone Map

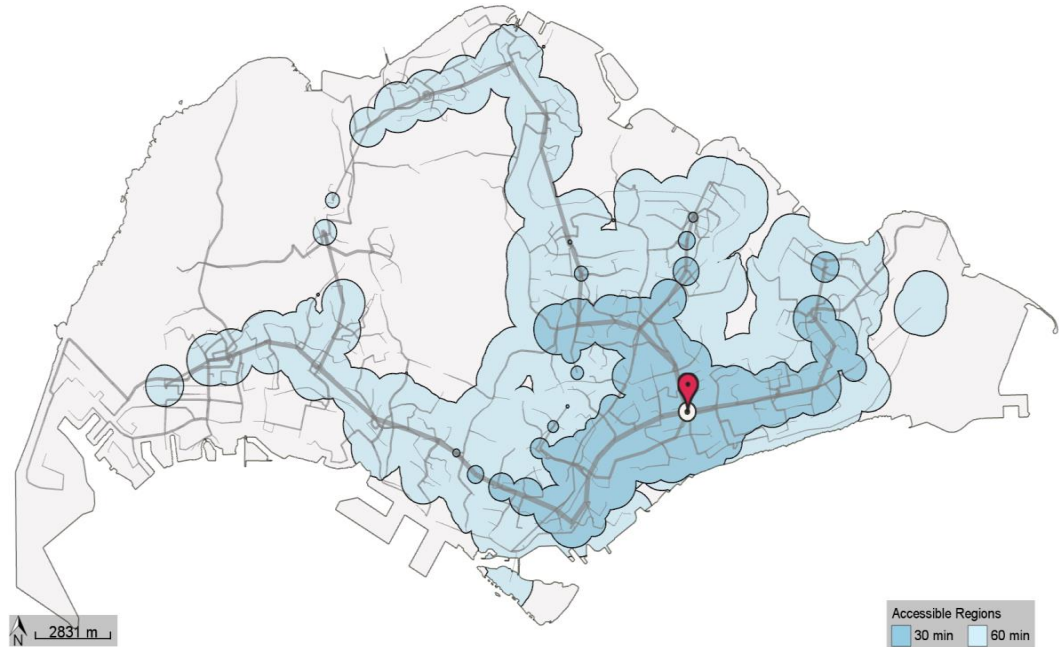
- We design a visual analytic system to explore PTS mobility with three visualization modules:

✓ **Isochrone Map**

✓ Isotime Flow Map

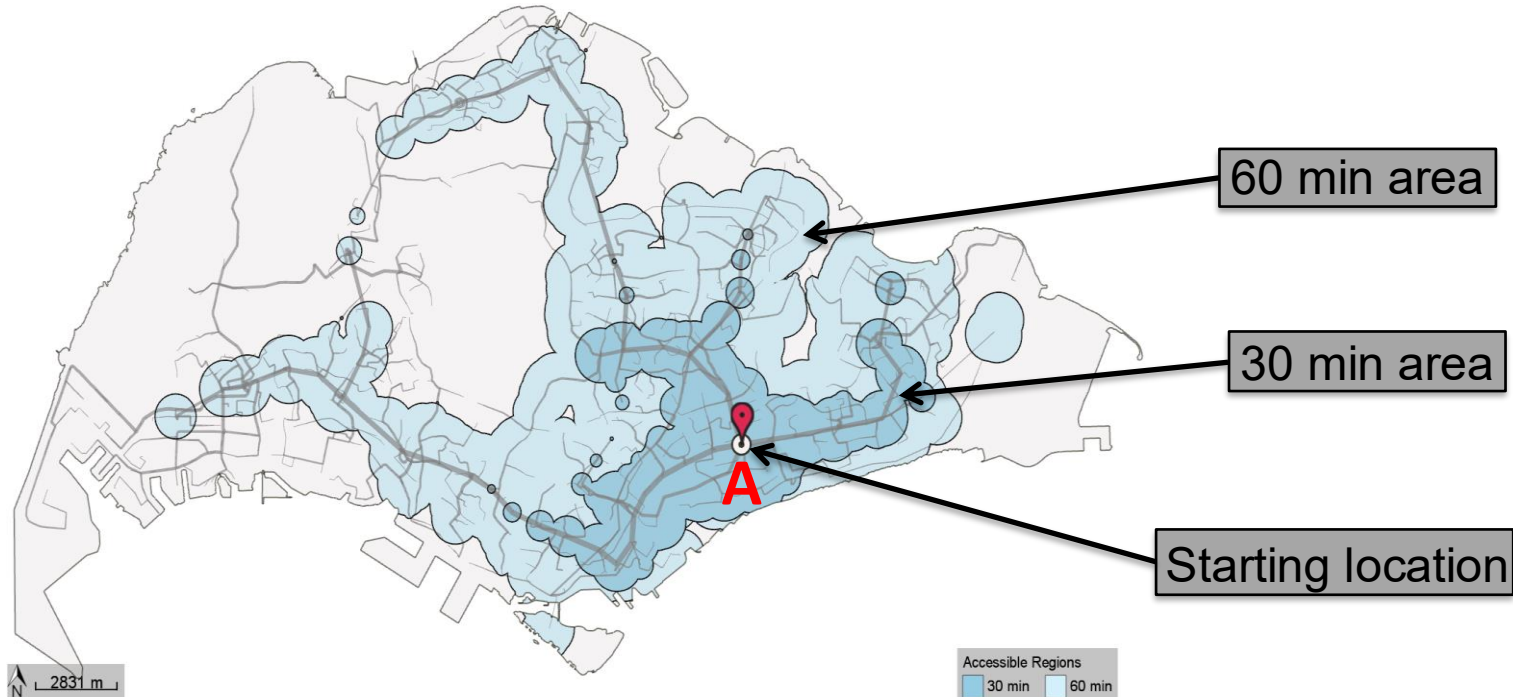
✓ OD-pair

Journey View

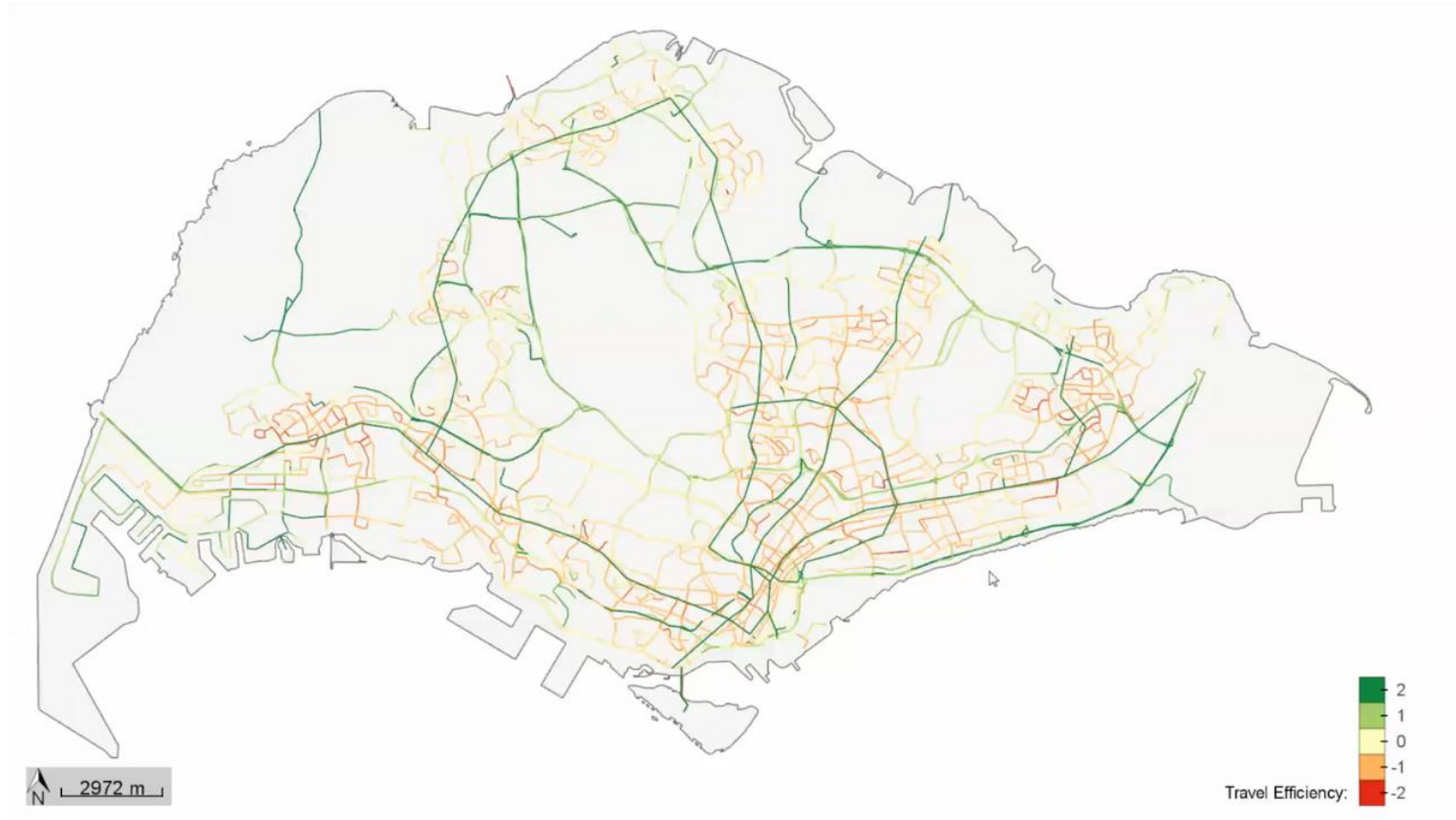


3.3 Isochrone Map

- ***Isochrone Map*** shows regions reachable from a given point A at time t_0 within a certain time duration (*Task 1*)



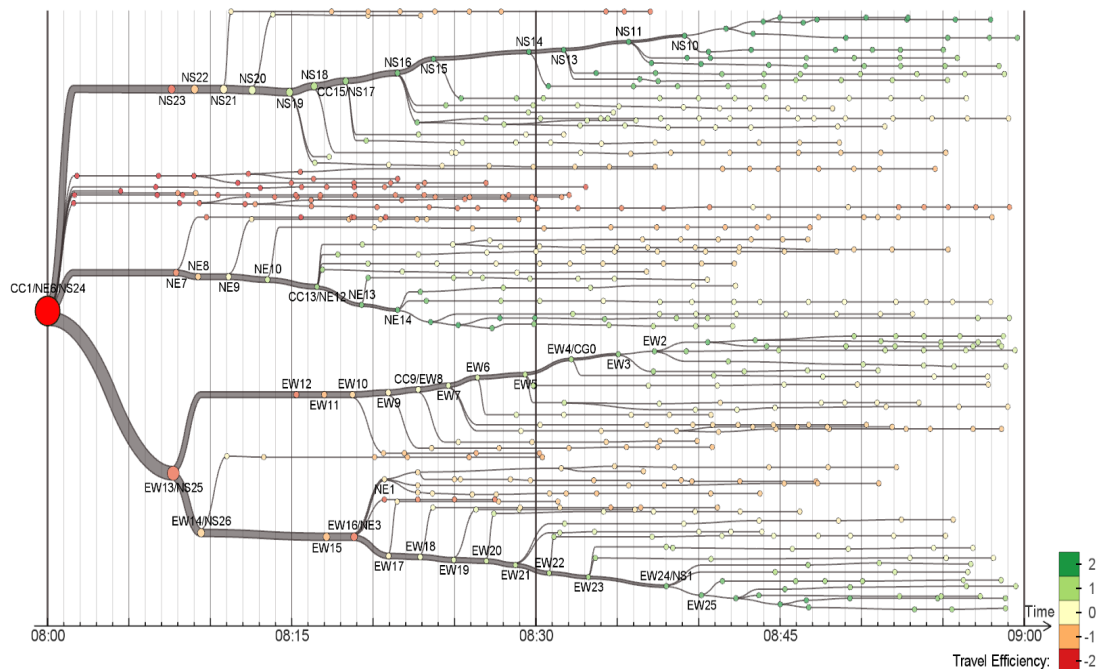
3.3 Isochrone Map



3.4 Isotime Flow Map

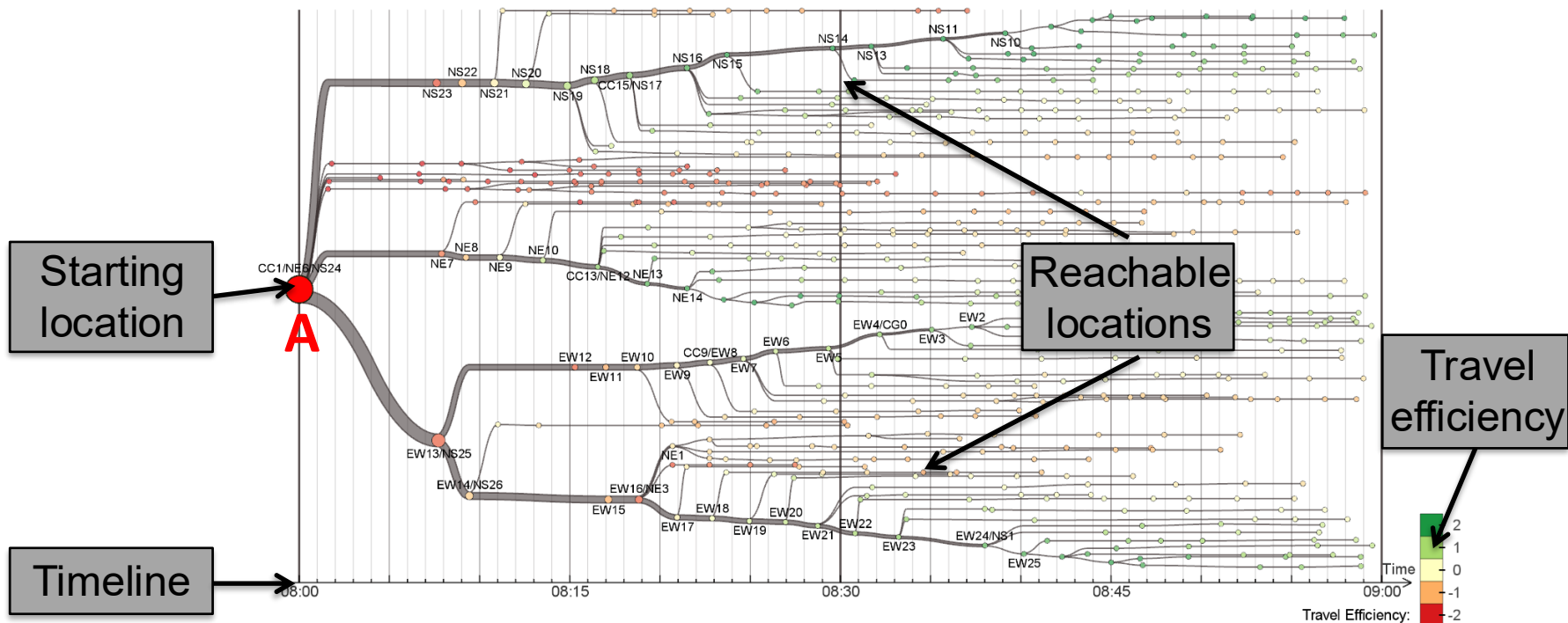
➤ We design a visual analytic system to explore PTS mobility with three visualization modules:

- ✓ Isochrone Map
- ✓ **Isotime Flow Map**
- ✓ OD-pair Journey View

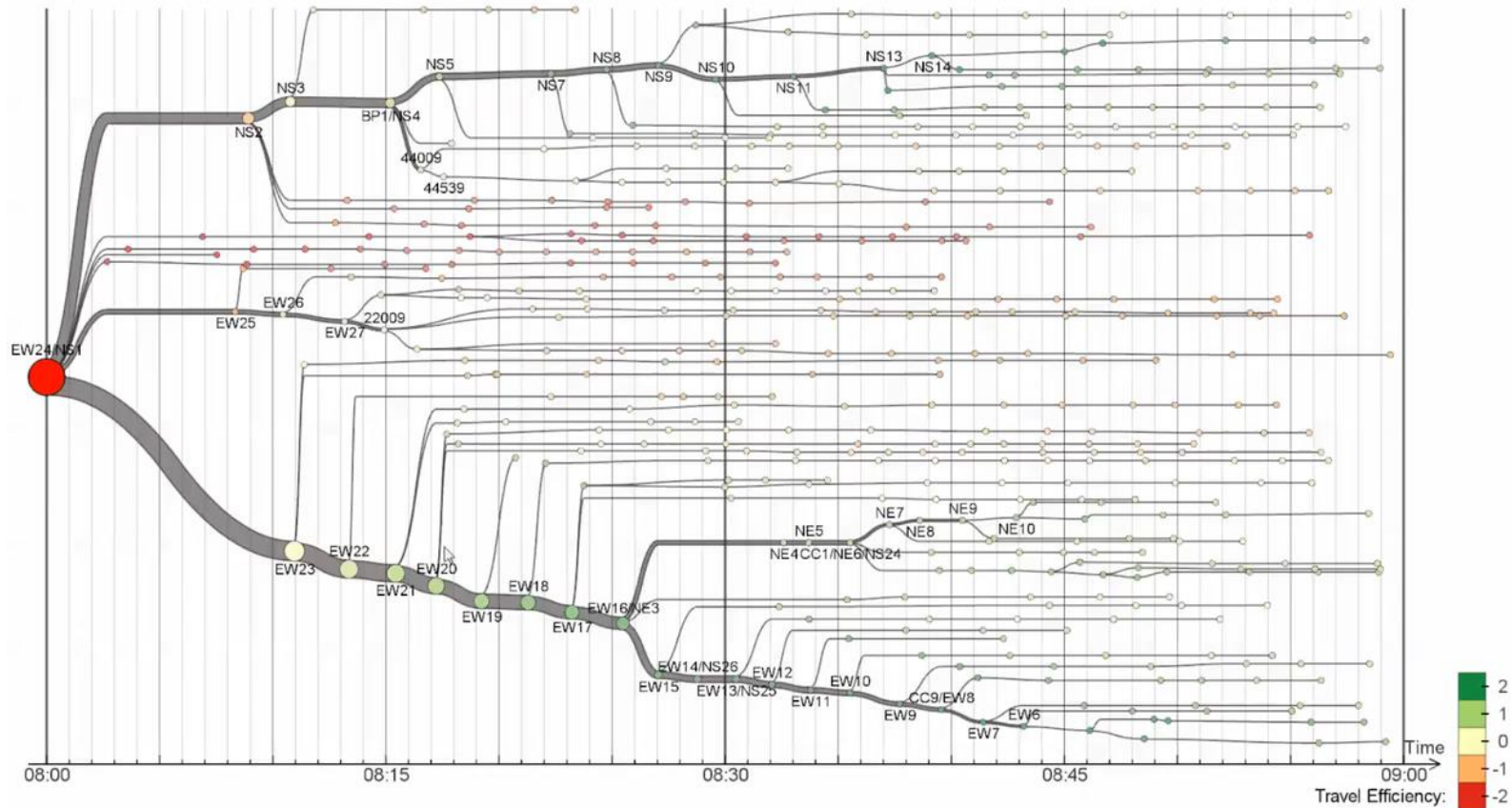


3.4 Isotime Flow Map

- **Isotime flow map** presents clear paths from *A* to all destinations, and facilitates exploration of travel time and efficiency (*Task 2 & 3*)



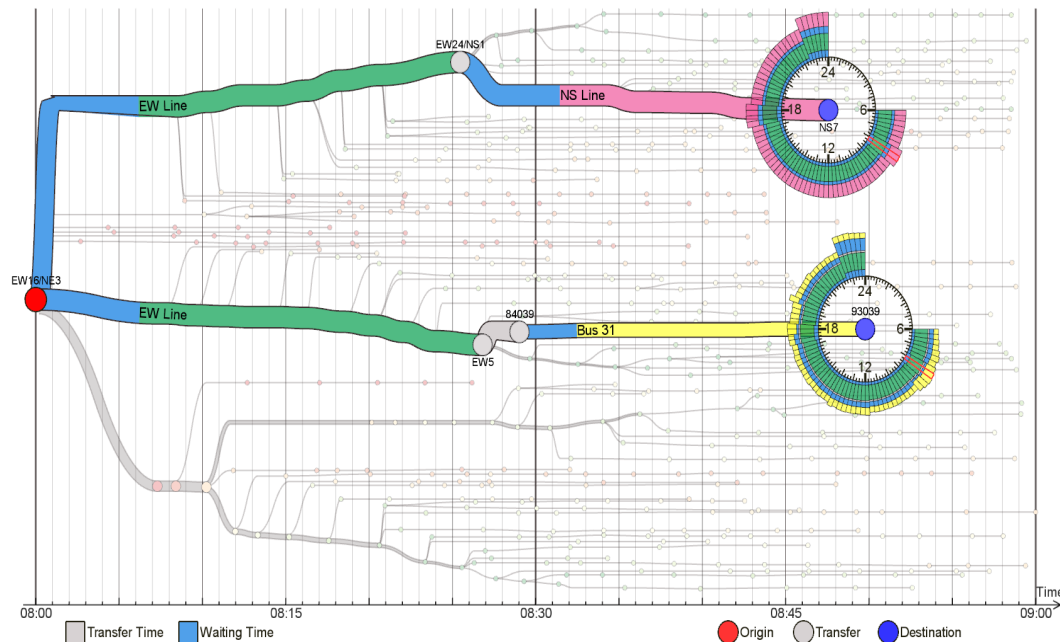
3.4 Isotime Flow Map



3.5 OD-pair Journey View

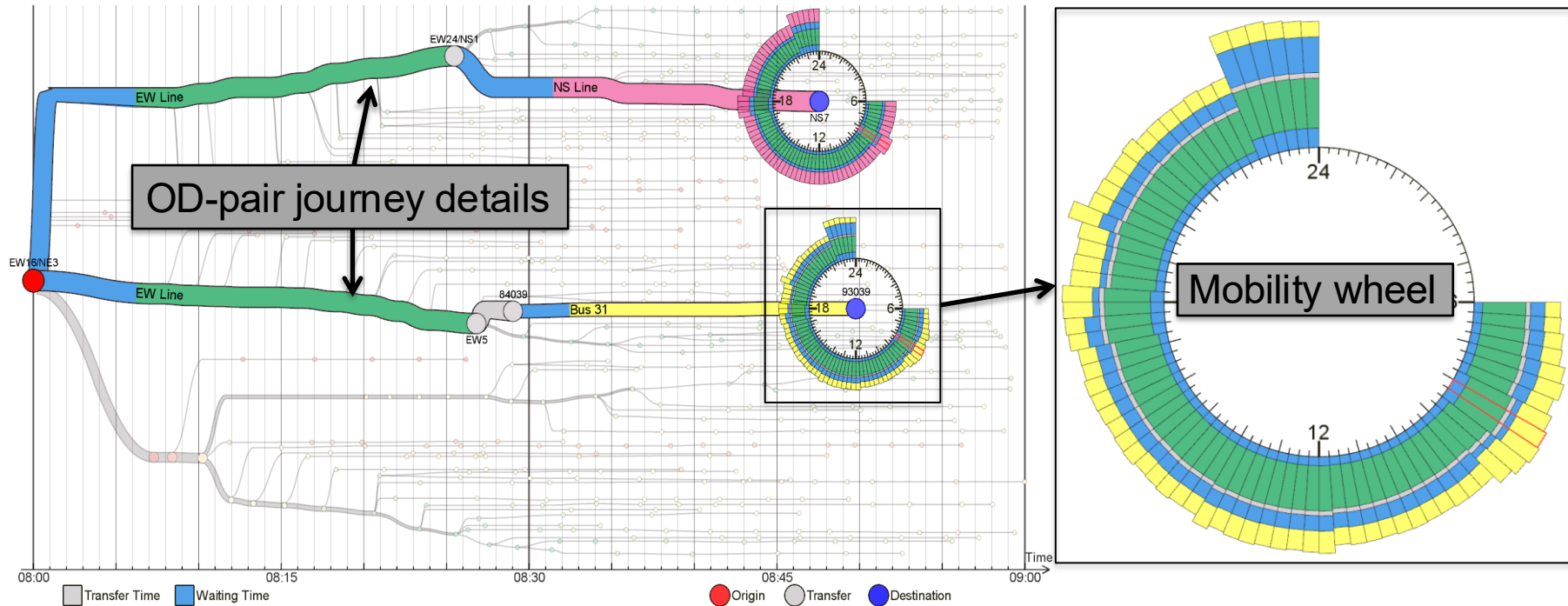
➤ We design a visual analytic system to explore PTS mobility with three visualization modules:

- ✓ Isochrone Map
- ✓ Isotime Flow Map
- ✓ **OD-pair Journey View**



3.5 OD-pair Journey View

- **OD-pair journey view** shows detailed mobility information, and their temporal variation along specific routes (*Task 4 & 5*)

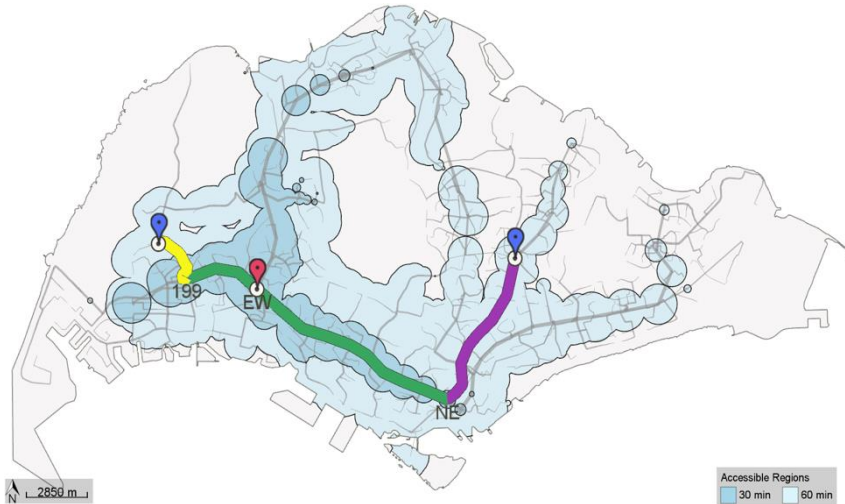


Contents

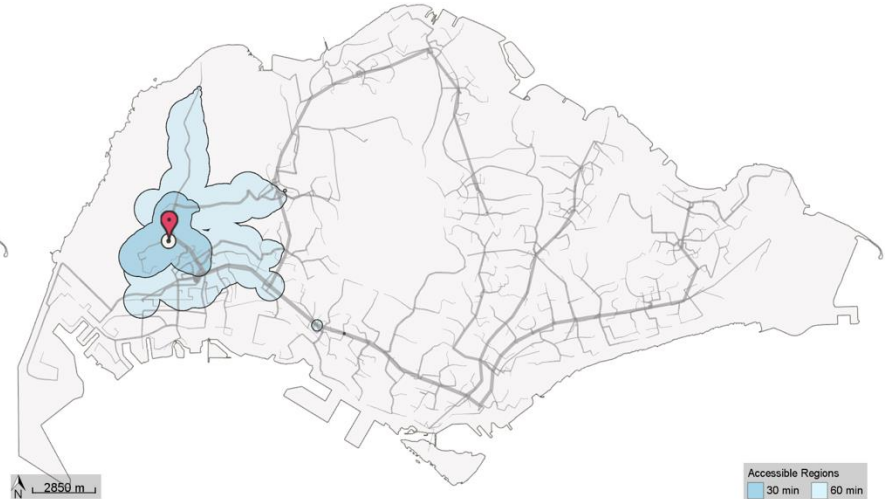
- 1. Introduction
- 2. Related Work
- 3. Our Visual Analytics System
- **4. Case Studies**
- 5. Conclusion and Future Work

4.1 Study 1: Spatial Variation

- Exploring the spatial variation of PTS mobility from two different origins at 8:00 A.M.



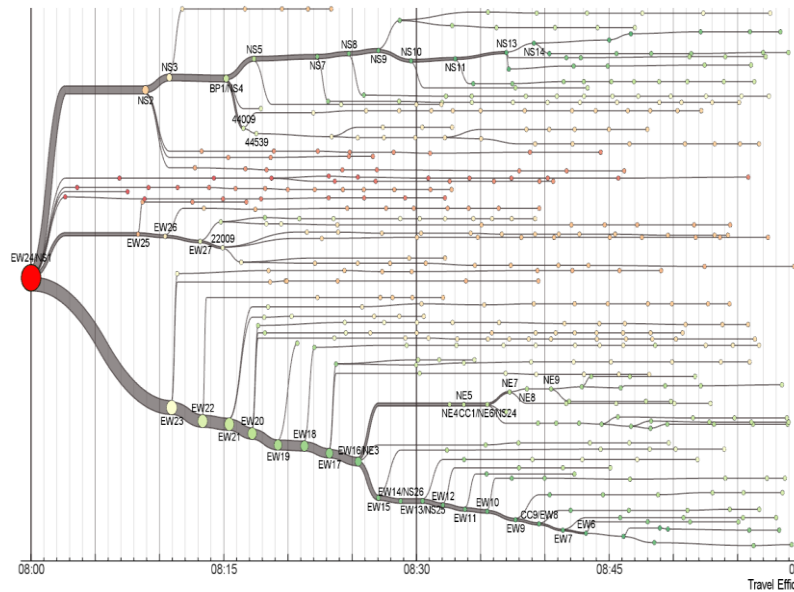
Jurong East MRT station



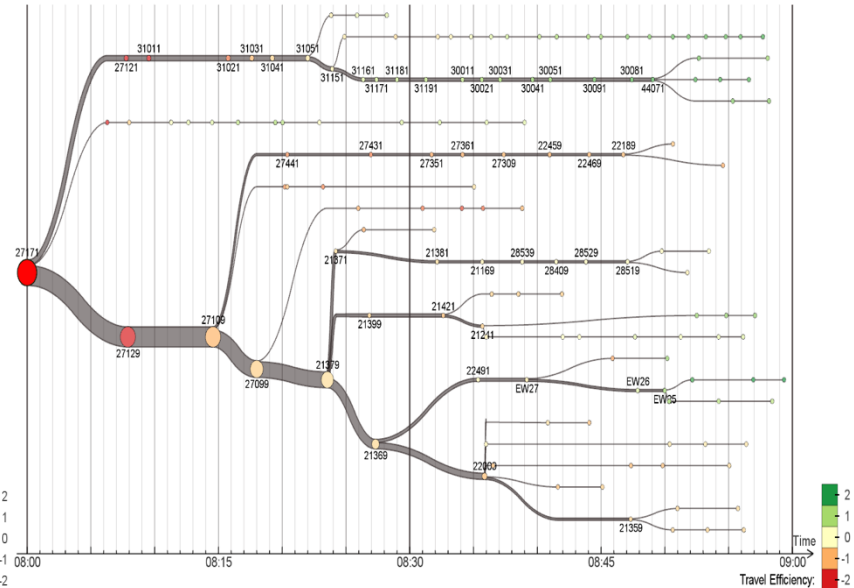
A remote bus stop at NTU

4.1 Study 1: Spatial Variation

- Exploring the spatial variation of PTS mobility from two different origins at 8:00 A.M.



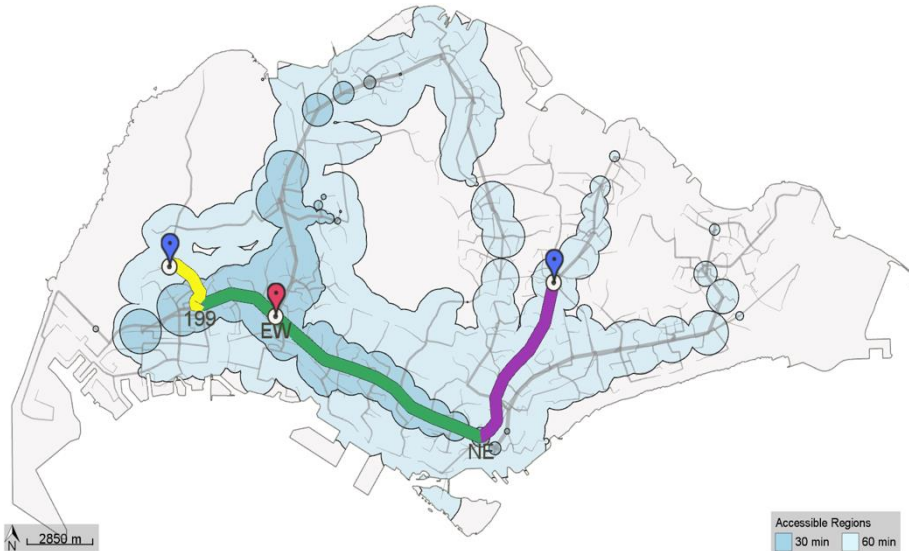
Jurong East MRT station



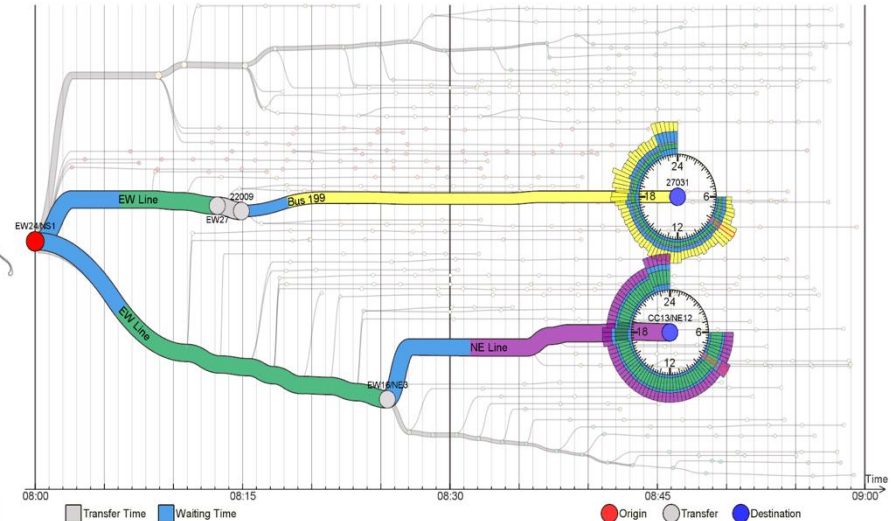
A remote bus stop at NTU

4.2 Study 2: Analyzing Mobility Factors

- Analyzing detailed mobility-related factors along different journeys starting from *Jurong East* to two destinations: 1) a *bus stop at NTU* (yellow), and 2) *Serangoon MRT station* (purple)



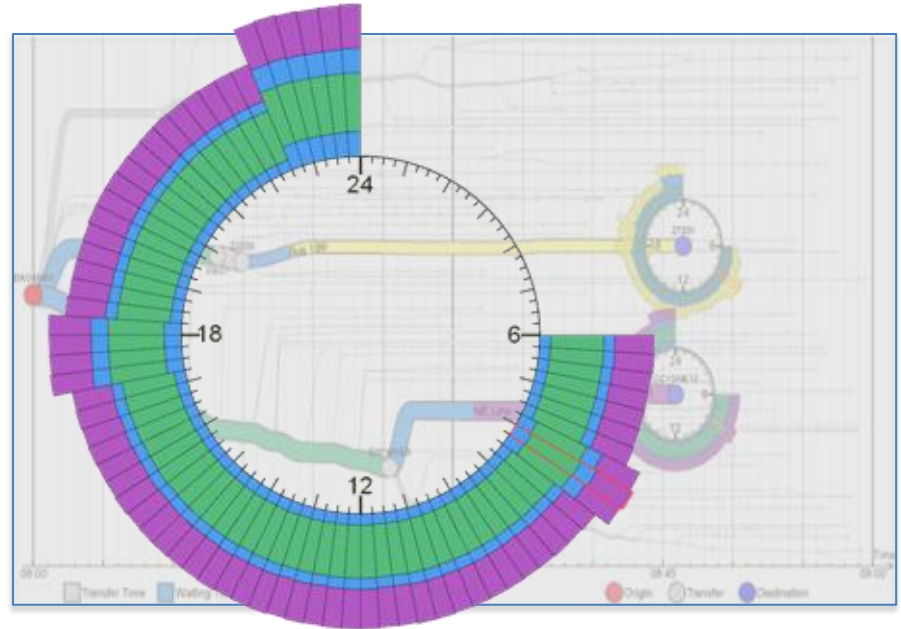
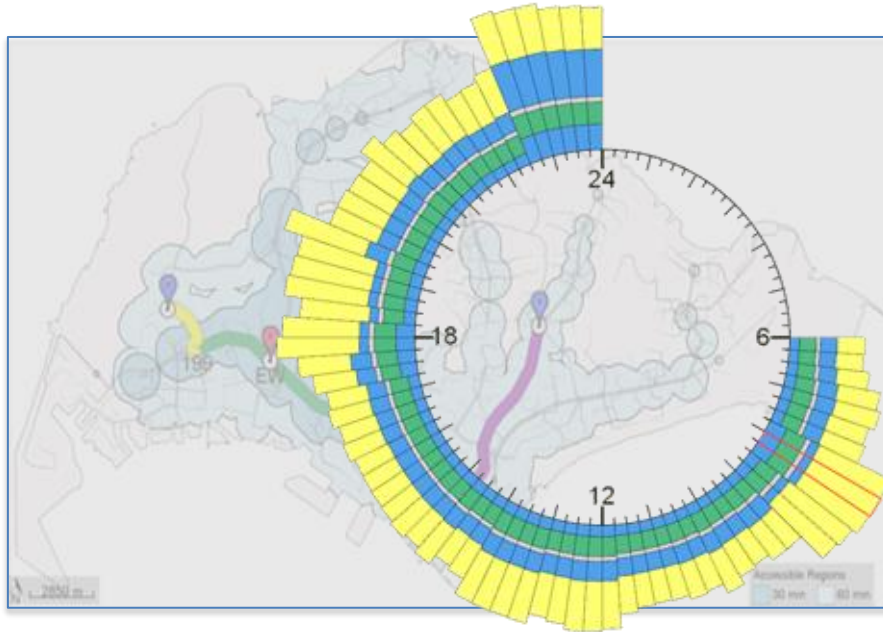
Two destinations are selected



The OD-pair journey view

4.8 Study 2: Analyzing Mobility Factors

- Zoomed views of the mobility wheels in the OD-pair journey view for presenting the round-the-clock mobility patterns



Contents

- 1. Introduction
- 2. Related Work
- 3. Our Visual Analytics System
- 4. Case Studies
- **5. Conclusion and Future Work**

5.1 Conclusion

- We explore and visualize PTS mobility:
 - ✓ We introduce and construct a **PTS mobility model** that characterizes the passenger mobility, and derive methods to estimate various mobility-related factors
 - ✓ We design an **integrated visual analytic interface** with three visualization modules: isochrone map view, isotime flow map view, and OD-pair journey view
 - ✓ We explore **two case studies with the transportation researchers**, and present their expert feedbacks on the interface design

5.2 Future Work

- We plan to derive **improved mobility factors estimation methods** by incorporating more real-world data into the estimation process
- We would like to explore **other transportation attributes** such as vehicle capacity, passenger composition (e.g., senior, student, and disabled), etc.
- We plan to extend our system for **real-time analysis of PTS mobility**, so that we may deliver adaptive journey planning for passengers

Thank You!

